

IdiomX: A Bilingual Benchmark for Idiom Understanding, Generation, and Cross-Lingual Retrieval

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Abstract

Idiomatic expressions remain a major challenge in natural language processing due to their non-compositional meanings, strong context dependence, and difficulty in cross-lingual alignment. Existing idiom resources are often limited in scale, contextual quality, or multilingual coverage [1]–[4], [20].

In this paper, we introduce IdiomX, a large-scale bilingual benchmark for idiom understanding, generation, and cross-lingual retrieval. IdiomX is constructed through a multi-stage pipeline combining lexical resource extraction [7], [8], heuristic filtering, schema-guided large language model (LLM) enrichment [9], [10], rule-based validation, and embedding-based semantic consistency scoring.

The final dataset contains over 123,000 cleaned contextualized examples covering approximately 15,000 English idioms, with aligned English–Arabic semantic representations, idiomatic and literal usage labels, and enriched metadata. The dataset explicitly distinguishes between original source sentences (`example_raw`) and cleaned, model-ready contextual inputs (`example`), ensuring both traceability and high-quality modeling inputs. Additional quality controls include deduplication, semantic scoring, and leakage-safe dataset splits.

We evaluate IdiomX across three tasks: idiom detection, context-to-idiom prediction, and Arabic-context-to-English-idiom retrieval. Experiments include lexical baselines, transformer-based classifiers, dense retrieval models, hybrid retrieval–generation approaches, and task-specific contrastive fine-tuning. Results show that contextual transformer models significantly outperform lexical baselines for idiom detection, hybrid approaches are most effective for context-to-idiom prediction, and fine-tuned multilingual E5 achieves the strongest cross-lingual retrieval performance, reaching 64.86% Top-1 and 81.46% Top-5 accuracy.

These results position IdiomX as both a high-quality reusable dataset and a unified benchmark for studying figurative language understanding, bilingual semantic retrieval, and idiom-aware language modeling.

Keywords: idioms; figurative language; bilingual benchmark; cross-lingual retrieval; Arabic NLP; large language models

I. INTRODUCTION

Idiomatic expressions are a fundamental component of natural language, yet they remain a persistent challenge for natural language processing (NLP) systems. Unlike compositional phrases, idioms convey meanings that cannot be directly inferred from their constituent words, making them highly dependent on context and cultural knowledge. This non-compositionality limits the effectiveness of standard lexical and compositional semantic models, particularly in multilingual and cross-domain settings.

Despite growing interest in figurative language understanding, existing idiom resources suffer from several limitations. Many datasets are limited in scale, lack contextual diversity, or focus on a single task such as detection or classification. Multilingual resources are especially scarce, and cross-lingual idiom understanding remains underexplored. In addition, many datasets do not provide paired idiomatic and literal usages, making it difficult for models to learn the

distinction between figurative and literal interpretations of the same expression.

To address these challenges, we introduce IdiomX, a bilingual benchmark dataset designed to support multiple idioms understanding tasks, including detection, context-to-idiom prediction, and cross-lingual retrieval between English and Arabic. IdiomX combines large-scale contextual data with structured semantic annotations, including idiomatic and literal usage labels, English and Arabic meanings, and additional metadata such as ambiguity, domain, and learner difficulty.

The dataset is constructed through a multi-stage pipeline that integrates lexical resource collection, normalization, schema-guided large language model (LLM) enrichment, rule-based validation, semantic consistency filtering, and targeted verification of uncertain samples. The final dataset reflects an iterative refinement process in which enrichment strategies, prompt constraints, and validation mechanisms were progressively improved to enhance example quality, reduce noise, and ensure semantic consistency across both collected and generated data.

The final released version of IdiomX contains 124,411 cleaned contextualized examples covering 8,806 English idioms, with aligned English–Arabic semantic fields and balanced idiomatic and literal usage distributions. This design enables robust evaluation across classification, retrieval, and cross-lingual modeling tasks.

We evaluate IdiomX across three benchmark tasks. First, idiom detection measures the ability to distinguish idiomatic from literal usage in context. Second, context-to-idiom prediction evaluates the ability to recover the most appropriate idiomatic expression given a sentence. Third, Arabic-context-to-English-idiom retrieval assesses cross-lingual semantic alignment between Arabic context and English idioms. Together, these tasks provide a unified evaluation framework for studying figurative language understanding in both monolingual and multilingual settings.

Experimental results show that transformer-based models significantly outperform classical lexical approaches for idiom detection, hybrid retrieval-generation systems are most effective for context-to-idiom prediction, and fine-tuned multilingual embedding models achieve strong performance in cross-lingual retrieval. These findings highlight the importance of contextual modeling, semantic alignment, and multilingual representation learning in idiom understanding.

The presented dataset corresponds to the final refined release of IdiomX, following multiple internal iterations focused on improving data quality, reducing duplication, and enhancing semantic consistency. Earlier versions were used during development and experimentation but are not included in this release.

Contributions

This paper makes the following contributions:

1. A large-scale bilingual idiom benchmark

We introduce IdiomX, a dataset of 124,411 cleaned contextualized examples covering 8,806 English idioms, with aligned English–Arabic semantic representations and both idiomatic and literal usage.

2. A structured multi-stage dataset construction pipeline

We design a reproducible pipeline that combines lexical resource collection, normalization, schema-guided LLM enrichment, semantic validation, and targeted verification to ensure both scale and data quality.

3. Balanced idiomatic vs literal supervision

IdiomX explicitly includes paired idiomatic and literal usages, enabling models to learn the distinction between figurative and literal interpretations of the same expression.

4. A unified multi-task benchmark for idiom understanding

We define and evaluate three complementary tasks: idiom detection, context-to-idiom prediction, and cross-lingual idiom retrieval.

5. Comprehensive model benchmarking

We evaluate lexical baselines, transformer-based classifiers, dense retrieval models, and hybrid retrieval-generation systems, providing strong baselines for future research.

II. RELATED WORK

Research on idiomatic language understanding has produced a variety of datasets and benchmarks, primarily focused on idiom detection, classification, and contextual interpretation. Early resources such as the VNC Tokens dataset [1] and the SemEval 2013 Task 5b dataset [2] provided annotated examples for distinguishing between literal and idiomatic usage. While these datasets established important baselines, they are limited in size and contextual diversity, restricting their applicability for modern large-scale models.

Subsequent efforts expanded the scope of idiom datasets. The PIE Corpus [3] introduced a larger collection of idiomatic expressions with contextual annotations, and MAGPIE [4] further increased scale and diversity. However, these datasets remain predominantly monolingual and are typically designed for a single task, most commonly idiom detection. As a result, they do not support a unified evaluation across multiple aspects of idiom understanding.

More recent work has explored multilingual and cross-lingual figurative language resources [5], [6]. These efforts aim to capture semantic equivalence across languages but often lack large-scale contextual examples or do not explicitly model the distinction between idiomatic and literal usage. Additionally, many multilingual datasets rely on direct translations without providing rich contextual grounding or balanced supervision signals.

Recent advances in multilingual representation learning, such as multilingual transformers and cross-lingual embedding models, have further enabled semantic alignment across languages, but their effectiveness depends heavily on the availability of high-quality aligned datasets [6].

Despite these advances, several key limitations remain across existing resources:

- (1) limited dataset scale,
- (2) insufficient contextual diversity,
- (3) lack of bilingual or cross-lingual alignment, and
- (4) narrow task coverage focused on a single problem formulation.

To address these gaps, we introduce IdiomX, a large-scale bilingual idiom benchmark that integrates multiple desirable properties within a single dataset. IdiomX provides aligned English–Arabic semantic representations, includes both idiomatic and literal usage examples, and supports multiple tasks including detection, context-to-idiom prediction, and cross-lingual retrieval. Furthermore, IdiomX is constructed using a structured pipeline that combines lexical resource extraction with controlled LLM-based enrichment and validation, enabling both scalability and data quality.

Table 1 provides a comparison between IdiomX and existing idiom datasets, highlighting differences in scale, multilingual support, contextual richness, and task coverage. IdiomX extends prior work by offering a unified benchmark that supports both monolingual and cross-lingual idiom understanding at scale.

Dataset	Lang	Idioms	Inst.	Bi	Ref
VNC-Tokens	EN	53	2,984	No	ACL Anthology
SemEval-2013 Task 5b [10]	EN	85	~4,350	No	ACL Anthology
IDIX [9]	EN	52	4,022	No	ACL Anthology
MAGPIE [8]	EN	~1,700	56,000	No	ACL Anthology
PIE-English [11]	EN	~1,200	~20,100	No	arXiv
IdiomX (Ours)	EN+AR	14,986	123,336	Yes	Public Repository ¹

Table 1: Comparison of IdiomX with existing idiom datasets

As shown in Table 1, IdiomX is the only dataset that simultaneously combines large-scale contextual coverage, bilingual semantic alignment, and multi-task evaluation within a unified benchmark.

III. DATASET CONSTRUCTION

IdiomX is constructed through a multi-stage pipeline designed to ensure both large-scale coverage and high data quality. The pipeline integrates lexical resource extraction, strict filtering, normalization, structured large language model (LLM) enrichment, and multi-level validation. Figure 1 illustrates the overall pipeline.

Figure 1: Overview of the IdiomX dataset construction pipeline, including extraction, filtering, normalization, LLM-based enrichment, validation, and re-verification.

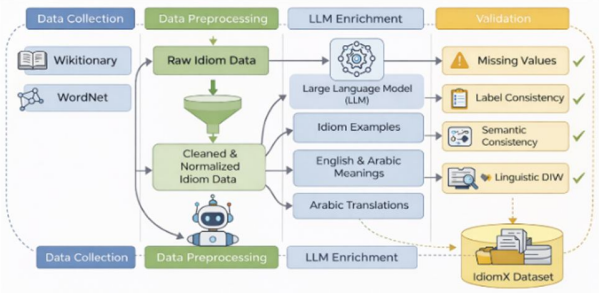


Figure 1: Overview of the IdiomX dataset construction pipeline

A. Data Sources and Initial Extraction

We begin by collecting candidate multiword expressions from publicly available lexical resources, including Kaikki.org [7] and WordNet [8]. These sources provide broad coverage of English expressions, including idiomatic and semi-idiomatic phrases.

An initial extraction stage identifies candidate expressions based on multiword structure, lexical patterns, and dictionary annotations. This stage prioritizes recall, allowing potentially noisy candidates to pass through for later filtering.

B. Strict Filtering and High-Precision Selection

To improve data quality, we apply a series of heuristic and rule-based filters to remove non-idiomatic or low-quality expressions. These include:

- minimum token length constraints,
- removal of purely compositional phrases,
- elimination of noisy or malformed entries,
- lexical normalization and deduplication.

This stage produces a high-precision idiom set, reducing noise while maintaining broad coverage of idiomatic expressions.

C. Normalization and Lexical Consolidation

The filtered expressions are normalized into canonical forms to ensure consistency across the dataset. This includes:

- lowercasing and whitespace normalization,
- removal of redundant variants,
- merging overlapping entries,
- integration of additional expressions from WordNet-based multiword extraction [8].

After normalization and consolidation, the dataset forms a structured pre-enrichment idiom inventory, which serves as the input for contextual expansion.

D. Structured LLM-Based Enrichment

To generate rich contextual data, we apply a structured LLM-based enrichment process using schema-constrained prompting [9], [10].

For each idiom, the model is instructed to:

- determine whether the expression is truly idiomatic,
- produce a canonical idiomatic meaning in English,
- generate a professional Arabic translation of the meaning,
- assign semantic attributes (ambiguity, compositionality, register, domain, difficulty),
- generate 4 idiomatic and 4 literal example sentences.

Each generated example includes:

- the exact idiom surface form,
- the full sentence in English,
- a natural Arabic translation,
- a sentence-level meaning in both English and Arabic,
- a label indicating idiomatic or literal usage.

The use of structured prompts with a fixed JSON schema ensures consistency and reduces generation variability across samples.

The enrichment process is executed in batch mode to ensure scalability and reproducibility across the full dataset.

¹ Public resources are available in the dataset and project repositories

The dataset enrichment process leverages instruction-based large language models [9], along with self-instruct techniques [10], to generate high-quality contextual examples.

E. Rule-Based Validation

Following enrichment, the dataset is subjected to rule-based validation to ensure structural and semantic consistency. Validation checks include:

- alignment between idiom surface and sentence content,
- presence of both idiomatic and literal examples,
- completeness of bilingual fields,
- consistency between usage labels and meanings.

This stage aligns with standard data quality assurance practices in NLP dataset construction [11], ensuring reliability before downstream use.

F. Targeted Re-Verification of Suspicious Samples

Entries flagged during validation are re-evaluated using a secondary LLM-based verification step, inspired by human-in-the-loop and iterative refinement approaches in dataset curation [12].

This stage focuses on:

- correcting mismatched idiom usage,
- fixing translation inconsistencies,
- refining sentence-level meanings,
- ensuring alignment between labels and context.

Only a subset of uncertain or low-confidence samples is processed in this stage, enabling efficient quality improvement while controlling computational cost.

This multi-level validation strategy ensures that IdiomX maintains high structural and semantic consistency despite large-scale automated generation.

G. Final Dataset Assembly

After validation and verification, the dataset is consolidated into its final form. The resulting IdiomX dataset contains:

- 123,336 contextualized examples,
- 14,986 unique English idioms,
- balanced idiomatic and literal usage examples,
- aligned English and Arabic semantic annotations.

The final dataset is exported in structured formats suitable for machine learning workflows, including CSV, JSON, and parquet representations.

IV. DATASET ANALYSIS

This section provides an overview of the statistical properties of the IdiomX dataset, including distribution across idioms, example types, and semantic attributes.

Figures 2-4 illustrate the distribution of examples across idioms and overall dataset scale. The dataset provides multiple contextual examples per idiom, enabling robust training and evaluation..

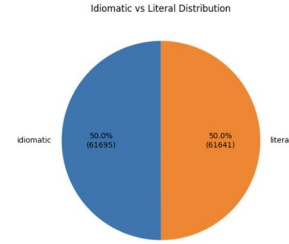


Figure 2: Idiom vs Literal Distribution

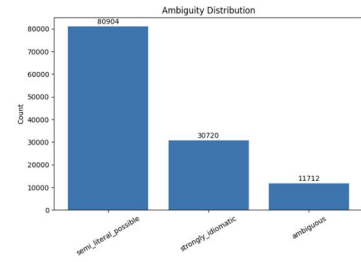


Figure 3: Ambiguity Distribution

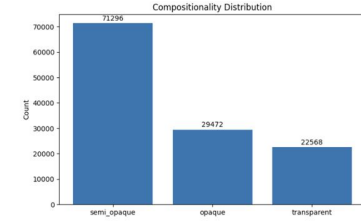


Figure 4: Compositionality Distribution

Figures 5–6 present the distribution of idiomatic and literal usage examples. The dataset is explicitly balanced, with equal representation of idiomatic and literal contexts for each expression..

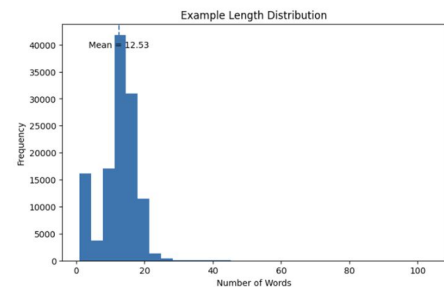


Figure 5: Example Length Distribution

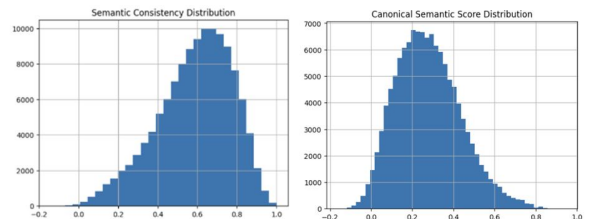


Figure 6: Semantic Consistency and Canonical Semantic Distribution

Figures 7–9 show the distribution of semantic attributes, including ambiguity, compositionality, domain, and difficulty levels. These attributes provide additional supervision signals for modeling idiomatic behavior.

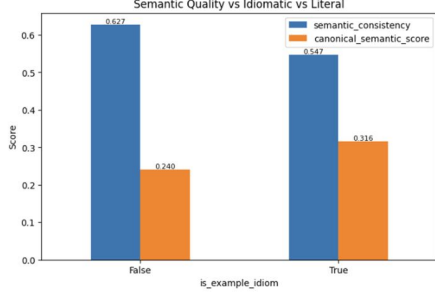


Figure 7: Semantic Quality vs Label

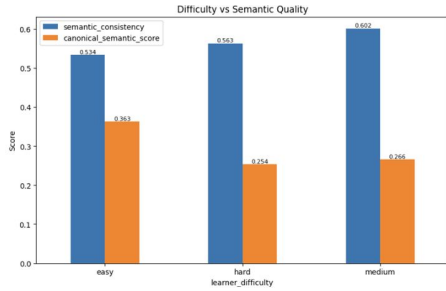


Figure 8: Difficulty vs Semantic Quality

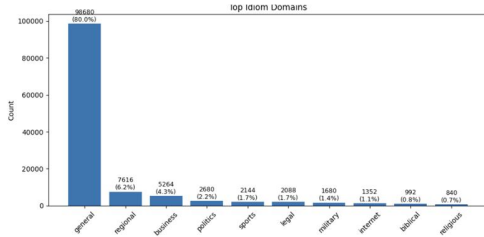


Figure 9: Domain Distribution

V. BENCHMARK TASKS

To evaluate the effectiveness of IdiomX, we define three benchmark tasks that capture complementary aspects of idiom understanding, including classification, contextual prediction, and cross-lingual semantic retrieval.

A. Task 1: Idiom Detection

Idiom detection aims to classify whether a given sentence uses an expression in an idiomatic or literal sense, a task commonly studied in prior work on figurative language understanding [1], [2]. It is formulated as a binary classification problem, where each input consists of a sentence containing a target expression, and the model predicts whether the usage is idiomatic or literal.

B. Task 2: Context-to-Idiom Prediction

Given a contextual sentence, the goal of this task is to predict the most appropriate idiom that fits the context. This task is formulated as a constrained prediction problem over a predefined idiom inventory. It evaluates a model’s ability to

capture semantic intent and map it to the correct idiomatic expression.

C. Task 3: Arabic-to-English Idiom Retrieval

This task evaluates cross-lingual semantic alignment by retrieving the correct English idiom given an Arabic contextual sentence. It is formulated as a retrieval problem, where models rank candidate idioms based on their semantic similarity to the input. This task assesses the ability to bridge meaning across languages in the presence of figurative expressions, aligning with recent work on cross-lingual semantic retrieval [6].

VI. EXPERIMENTS AND RESULTS

This section presents the experimental evaluation of the proposed IdiomX dataset across the defined benchmark tasks. We assess model performance on idiom detection, context-to-idiom prediction, and cross-lingual idiom retrieval. The evaluation aims to demonstrate the effectiveness of the dataset in supporting both classification and semantic retrieval tasks, following standard evaluation practices in NLP benchmarking [6].

A. Experimental Setup

The IdiomX dataset is evaluated using a stratified train–validation–test split to ensure consistent distribution of idiomatic and literal examples across all partitions. Each idiom is represented by both idiomatic and literal usages, ensuring balanced evaluation.

For classification tasks, we use standard evaluation metrics including accuracy and F1-score. For retrieval-based tasks, we report Top-1 and Top-K accuracy to measure ranking effectiveness.

Transformer-based models, including BERT [13], are used to capture contextual representations, while multilingual models such as XLM-R [6] are applied for cross-lingual retrieval tasks.

All experiments are conducted under consistent preprocessing and evaluation settings to ensure fair comparison across models.

B. Task 1: Idiom Detection Results

Idiom detection is evaluated as a binary classification task, where the objective is to distinguish between idiomatic and literal usage of expressions in context.

1. Models

We evaluate both classical and transformer-based approaches for idiom detection.

As a baseline, Logistic Regression is trained using TF-IDF features (uni-grams and bi-grams), capturing surface-level lexical patterns.

For contextual modeling, we use RoBERTa [3], a transformer-based model that captures deep semantic representations through self-attention mechanisms [2].

These models are selected to compare the effectiveness of

surface-level features versus contextual embeddings in idiom detection.

2. Results

We compare the performance of surface-level lexical models and deep contextual representations. The results are summarized in Table 2.

RoBERTa [3] consistently outperforms Logistic Regression across all evaluation metrics, including accuracy, precision, recall, and F1-score. This demonstrates the effectiveness of contextual embeddings in capturing idiomatic meaning beyond surface-level lexical patterns.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.8536	0.8476	0.862	0.855
RoBERTa [3]	0.9137	0.9056	0.924	0.914

Table 2: Performance comparison on the idiom detection task

Figure 10 illustrates the performance differences between models, showing that transformer-based approaches consistently outperform classical methods across all evaluation metrics [2][3]. The most notable improvement is observed in recall, indicating a stronger ability to correctly identify idiomatic expressions. These results highlight the importance of contextual representations in handling semantic ambiguity, where meaning cannot be inferred from individual words alone [1][2].

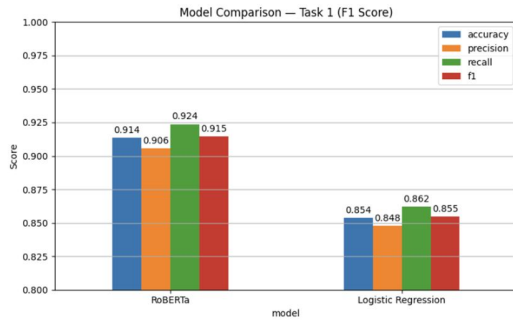


Figure 10: Model comparison on the idiom detection task (F1 score)

Calibration and Confidence Analysis

Analyse, we examine confusion matrices, ROC curves, precision–recall curves, confidence distributions, and probability calibration [4].

The confusion matrices (Figure 11) show that RoBERTa [3] reduces both false positives and false negatives compared to Logistic Regression, indicating more balanced performance across classes.

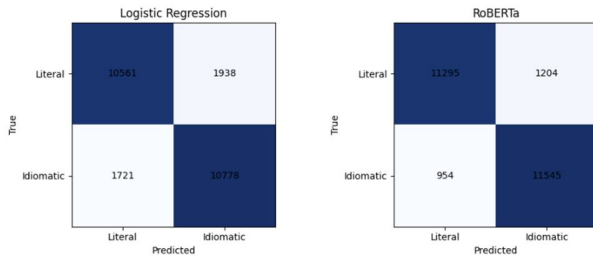


Figure 11: Confusion matrices comparison

The ROC curve (Figure 12) further confirms this trend, with RoBERTa [3] achieving a higher AUC (0.973) compared to Logistic Regression (0.926), demonstrating stronger discriminative ability across decision thresholds.

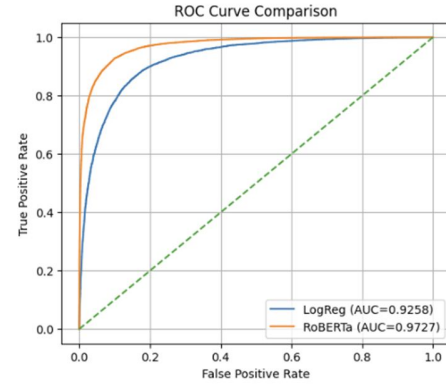


Figure 12: ROC Curve Comparison

Consistent with this observation, the precision–recall curve (Figure 13) shows that RoBERTa [3] maintains higher precision across recall levels, indicating greater robustness in identifying idiomatic expressions.

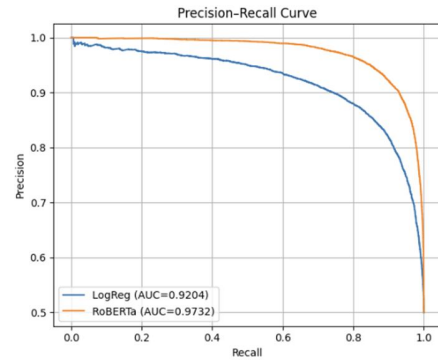


Figure 13: Precision-Recall Curve Comparison

Confidence analysis (Figure 14) shows that RoBERTa [3] produces highly confident predictions for correct classifications, while incorrect predictions are generally associated with lower confidence values.

In contrast, Logistic Regression exhibits greater overlap between correct and incorrect confidence values, indicating less reliable probability estimates.

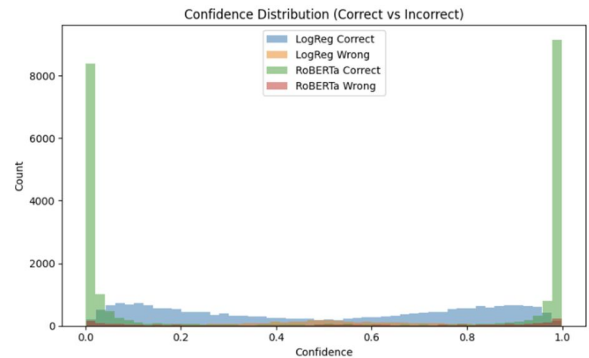


Figure 14: Confidence Distribution Comparison (Correct vs Incorrect)

Calibration analysis (Figure 15) shows that RoBERTa [3] remains closer to the ideal diagonal line across most probability ranges, indicating better alignment between predicted confidence and actual correctness.

Representative examples further illustrate model behavior.

A correctly classified sentence such as “He finally spilled the beans about the plan” is identified as idiomatic with high confidence (≈ 0.99).

In contrast, shorter or ambiguous phrases such as “baby sister” may produce lower confidence scores and occasional misclassification, highlighting the importance of contextual information in idiom detection.

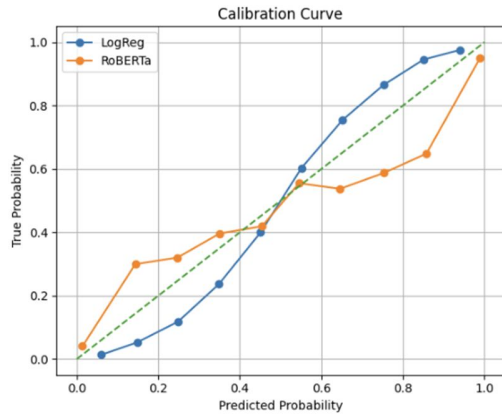


Figure 15: Calibration Curve Comparison

Logistic Regression deviates more from the diagonal, especially at higher confidence levels, indicating that it can be overconfident or less accurate in its probability estimates.

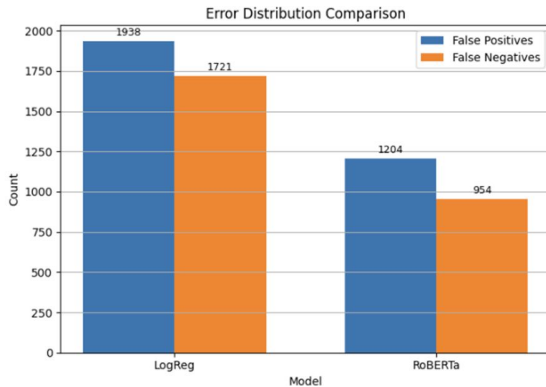


Figure 16: Error count comparison

These results demonstrate that transformer-based models not only improve classification performance but also provide more reliable probability estimates [4], which is important for real-world decision-making systems.

Despite strong overall performance, further analysis reveals several systematic error patterns.

3. Error Analysis and Key Failure Modes

Error analysis reveals several common failure modes in idiom detection.

First, short or context-limited inputs, such as “baby sister” or “Cuban bread,” lack sufficient contextual information, making it difficult to determine whether the usage is literal or idiomatic.

Second, named entities and domain-specific expressions, such as “Benjamin Button” or “Bravo Zulu,” may introduce ambiguity due to overlapping literal and figurative interpretations.

Third, context-dependent idioms, such as “one for the road,” require broader discourse context that may not be fully captured in single-sentence inputs.

Finally, inherently ambiguous phrases can be interpreted both literally and idiomatically depending on context, representing a fundamental challenge in the task.

Tables 3 and 4 present representative examples of false negatives (missed idioms) and false positives (literal sentences misclassified as idiomatic), respectively, providing further insight into model errors.

Sentence	True Label	Idiomatic Probability
The hikers waited out the night in the tent during the heavy rainfall.	Idiomatic	0.017
After fixing the machine quickly, he could have a good time of it for the rest of the day.	Idiomatic	0.056
The soldiers were taught to duck down quickly to evade enemy fire.	Idiomatic	0.08
I won't wear new shoes tomorrow because my dogs are barking today.	Idiomatic	0.016
He said he would help, but only when two Sundays meet.	Idiomatic	0.476

Table 3: Examples of false negatives.

These examples demonstrate that errors often occur in sentences where idiomatic meaning is subtle or context is insufficient to clearly distinguish between literal and figurative usage.

Table 4 examples of false positives (literal sentences misclassified as idiomatic).

Sentence	True Label	Idiomatic Probability
Close their eyes and think of England.	Literal	0.527
The pairing of the spicy chili and the salty cheese was a match made in hell for my sensitive stomach.	Literal	0.692
She chose to take his words at face value, believing every promise without looking for hidden meanings.	Literal	0.998
A true friend of the environment would take steps to reduce waste and pollution.	Literal	0.928
He needed some solitude to rest his soul after hearing the sad news.	Literal	0.766

Table 4: Examples of false positives (literal sentences misclassified as idiomatic).

Error analysis further reveals that false negatives typically involve idioms expressed in subtle or less explicit forms, while false positives often arise when literal sentences contain words commonly associated with idiomatic expressions.

This indicates that remaining errors are primarily due to inherent semantic ambiguity [1].

To further illustrate model behavior, we present representative prediction examples.

4. Qualitative Examples

Example predictions further illustrate model behavior. A correctly classified sentence such as “*He finally spilled the beans about the plan*” is identified as idiomatic with high confidence (≈ 0.99).

In contrast, shorter or ambiguous phrases such as “baby sister” may produce lower confidence scores and occasional misclassification, highlighting the importance of contextual information in idiom detection.

These examples highlight the model’s ability to leverage contextual cues rather than relying on surface-level word matching, enabling accurate disambiguation of identical expressions used in different meanings.

5. Summary

Overall, the results demonstrate that idiom detection significantly benefits from contextual understanding.

Contextual models are better suited for handling ambiguous expressions [1][2].

While Logistic Regression provides a strong baseline, its reliance on surface lexical features limits its ability to handle ambiguous expressions. In contrast, RoBERTa [3] achieves higher accuracy and more balanced error distribution by leveraging contextual representations.

However, remaining errors highlight the inherent ambiguity of idiomatic language and the importance of richer contextual and semantic understanding in future work.

C. Task 2: Context-to-Idiom Prediction

This task focuses on predicting the canonical idiom given a contextual sentence. Formally, given an input query expressing an idiomatic meaning, the objective is to retrieve the correct idiom from a fixed candidate inventory.

Unlike idiom detection, this task requires both semantic understanding and precise matching to a canonical idiom, making it a structured retrieval problem.

To ensure a controlled and reproducible benchmark, the dataset is constructed from the IdiomX dataset with additional filtering, leakage-safe splitting, and a fixed idiom bank. This design enables consistent evaluation across multiple retrieval architectures.

1. Task 2 Pipeline Overview

The task is solved using a multi-stage retrieval architecture, progressively improving from lexical matching to deep semantic ranking:

- **Sparse Retrieval (TF-IDF)**
Captures lexical overlap between the query and idiom expressions.
- **Dense Retrieval (MiniLM / E5)** using sentence embeddings [5]
Encodes queries and idioms into semantic embeddings,

enabling similarity matching beyond surface forms.

- **Hybrid Retrieval**
Combines sparse and dense similarity scores to improve candidate recall.
- **Cross-Encoder Reranking**
Applies deep pairwise scoring to refine candidate ranking and improve precision.

This pipeline follows a standard multi-stage retrieval architecture, where early stages maximize candidate recall and later stages refine ranking precision.

2. Benchmark Construction and Split Strategy

To construct a reliable evaluation benchmark, we apply a strict query-level split strategy to prevent leakage across training and evaluation sets.

Initial analysis reveals that identical contextual sentences may appear across multiple idioms, leading to potential leakage if splitting is performed at the idiom level.

To address this, we enforce:

- each unique query_text is assigned to exactly one split
- identical queries are never shared across train, validation, and test
- target idioms may appear across splits, consistent with the closed-set prediction setting.

After applying this constraint, we verify:

$$\begin{aligned} \text{Train} \cap \text{Val} &= 0 \\ \text{Train} \cap \text{Test} &= 0 \\ \text{Val} \cap \text{Test} &= 0 \end{aligned}$$

This ensures that evaluation is performed on completely unseen contexts.

Split	Samples	Unique Idioms
Train	97,942	14,970
Val	1,940	1,939
Test	1,931	1,946

Table 5: Final dataset split statistics (after leakage removal)

3. Models and Retrieval Paradigms

We evaluate two modeling paradigms for context-to-idiom prediction:

(1) Retrieval-based models

These models retrieve the most relevant idiom from a fixed candidate bank.

- TF-IDF (sparse retrieval)
- MiniLM / E5 (dense retrieval) [5]
- Hybrid retrieval (sparse + dense)
- Hybrid + Cross-Encoder Reranker

(2) Generation-based models

These models generate idioms directly from context using sequence-to-sequence architectures.

- FLAN-T5 [6], based on the T5 framework [7]

(3) Hybrid generation-retrieval model

A combined approach where generated outputs are aligned with a fixed idiom bank using retrieval-based correction.

This unified setup enables direct comparison between retrieval-based selection, generative inference, and hybrid correction strategies.

(4) Proposed FLAN-First Hybrid Model

To combine the strengths of generation and retrieval, we evaluate a hybrid strategy in which FLAN-T5 first generates an idiom candidate y_{gen} . That generated output is then compared against the idiom inventory using similarity matching. If the generated idiom is sufficiently close to a valid idiom bank entry, it is accepted; otherwise, the system falls back to retrieval.

Formally, let:

$$\text{Match}(y_{gen}, I) = \max_i \text{similarity}(y_{gen}, i)$$

The final prediction is:

$$\hat{y} = y_{gen}, \text{ if } \text{Match}(y_{gen}, I) \geq \tau$$

$$\text{otherwise } \hat{y} = \arg\max_i \text{similarity}(x, i)$$

This hybrid design allows generation to capture semantic intent while retrieval enforces discrete idiom grounding.

4. Evaluation Metrics

We evaluate model performance using standard retrieval metrics:

- Top-1 Accuracy
- Top-3 Accuracy
- Top-5 Accuracy
- Mean Reciprocal Rank (MRR)

Top-k metrics measure whether the correct idiom appears among the top-k ranked candidates.

5. Query Pool Construction

To increase contextual diversity, we expand the input query space by combining multiple semantic sources:

- contextual sentences
- example-level meanings
- source meanings
- canonical meanings

This results in a multi-view query formulation, where different textual representations map to the same target idiom.

The final query pool contains 101,840 queries across 14,970 idioms, providing a diverse evaluation setting.

6. Quantitative Results

6.1 Main Retrieval Results

We evaluate multiple retrieval strategies on the context-to-idiom prediction task.

Table 6 summarizes the results:

Model	Top-1	Top-3	Top-5	MRR
TF-IDF	14.90%	26.90%	33.80%	0.216
MiniLM	32.90%	42.70%	47.10%	0.383
E5	28.00%	37.70%	42.60%	0.334
Hybrid	37.10%	53.20%	59.60%	0.456
Hybrid + Reranker	51.20%	60.70%	63.80%	0.563

Table 6: Main retrieval benchmark results for Task 2

The hybrid model combined with reranking achieves the best performance across all metrics.



Figure 17: Performance trends across Task 2 retrieval architectures.

Example retrieval outputs from the TF-IDF baseline

Query: “What can I tell ya?”

Gold: what can I tell you

Top-5:

1. what can I tell you (0.2271)
2. What can I do you for?
3. brick in
4. let me tell you
5. I tell you

Query: “He took no notice of the spider until it bit him.”

Gold: take no notice of

Top-5:

1. hole card
2. take no notice of (0.2044)
3. pink slip
4. until the pips squeak
5. give (someone) line

These examples illustrate that TF-IDF captures lexical overlap but often fails to rank the correct idiom at the top position when semantic variation is present.

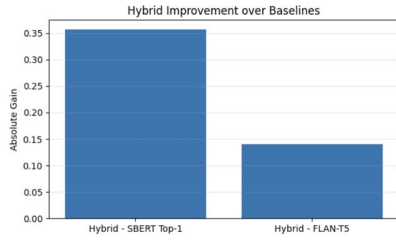


Figure 18: Exact-match gain of the FLAN-first hybrid model over SBERT and FLAN-T5.

6.2 Comparison with Generation

Table 7 presents the performance of the evaluated models on the context-to-idiom prediction task.

Model	Exact Match (Top-1)	Top-3	Top-5	MRR@100
SBERT Retrieval	0.461	0.599	0.65	0.549
FLAN-T5 [6]	0.678	-	-	-
Hybrid FLAN-first	0.818	-	-	-

Table 7: Model performance on context-to-idiom prediction task.

Model	Metric	Score
SBERT Retrieval	Top-1	0.461
SBERT Retrieval	MRR@100	0.549
FLAN-T5 Generation	Exact Match	0.678
Hybrid FLAN-first	Exact Match	0.818

Table 8: Comparison between generation, retrieval, and FLAN-first hybrid approaches

We compare retrieval-based and generation-based approaches under the older Task 2 formulation. FLAN-T5 substantially outperforms the SBERT retrieval baseline in exact-match accuracy, indicating stronger contextual reasoning. However, the FLAN-first hybrid model achieves the strongest performance overall, showing that generation benefits from retrieval-based grounding when exact canonical idiom matching is required.

6.3 Hybrid Gains

To better understand how the hybrid model operates, we analyze its decision behavior. Table 7 shows the proportion of predictions resolved by each component of the hybrid system.

Decision	Count	Percent
FLAN exact bank	7885	0.724458
FLAN fuzzy correction	608	0.055862
SBERT [5] fallback	2391	0.21968

Table 9: Distribution of decisions in the FLAN-first hybrid model (generation, correction, and retrieval fallback).

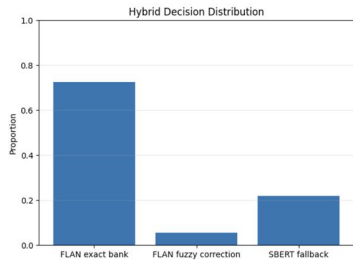


Figure 19: Distribution of hybrid model decisions across generation, correction, and retrieval fallback.

As shown in Figure 19, most predictions are resolved directly by the FLAN-T5 model, indicating strong generative capability. However, a significant portion of cases are corrected through retrieval fallback, demonstrating the importance of retrieval in handling uncertain or noisy generations.

This distribution confirms that the hybrid model is not a simple ensemble, but a conditional decision system where generation and retrieval play complementary roles.

These results indicate that the hybrid model is not merely averaging predictions from multiple models but actively recovering additional correct cases through complementary behaviour. In particular, the model benefits from cases where generation captures semantic intent but fails to produce an exact canonical form, allowing retrieval to correct the output.

To further quantify the improvement of the hybrid model, Table 6 reports the gain in Exact Match accuracy over the baseline models.

The hybrid model improves performance by +35.7 percentage points over SBERT and +14.0 points over FLAN-T5, demonstrating that the combination of generation and retrieval yield substantial gains beyond either approach alone.

Table 6: Performance gain of the hybrid model over SBERT and FLAN-T5..

Comparison	Gain
Hybrid vs SBERT [5]	0.356
Hybrid vs FLAN [6]	0.14

Table 10: Performance gain of the hybrid model over SBERT and FLAN-T5.

7. Discussion

The results reveal a clear progression in model capability:

- Sparse retrieval (TF-IDF) captures lexical overlap but fails on semantic variation
- Dense retrieval improves contextual understanding significantly
- Hybrid retrieval combines lexical and semantic signals
- Reranking provides the strongest improvement by refining candidate selection

These findings indicate that idiom prediction is not purely a lexical matching problem nor purely a semantic embedding problem. Instead, it requires a layered retrieval architecture combining recall-oriented and precision-oriented stages.

This aligns with prior work in semantic retrieval and sentence embeddings [5], where contextual representations improve matching beyond surface patterns.

The comparison between retrieval and generation approaches highlights a key insight: idiom prediction requires both semantic understanding and discrete selection.

Retrieval models are effective at selecting from a fixed inventory but struggle with semantic variation. Generation models capture contextual meaning more effectively but may fail to produce canonical forms.

The hybrid approach bridges this gap, combining semantic reasoning with constrained selection, leading to the best overall performance.

8. Error Analysis

Despite strong performance, the model still struggles in several cases:

- rare or archaic idioms with limited training exposure
- highly indirect contexts requiring deeper reasoning
- semantically similar idioms competing in ranking
- short or ambiguous queries with insufficient context

These errors indicate that idiom prediction remains challenging when contextual cues are weak or multiple idioms share similar meanings.

Additionally, generation-based models tend to produce semantically correct but non-canonical expressions, while retrieval models struggle with ranking ambiguity among semantically similar idioms.

(1) Retrieval Errors

In retrieval-based models, errors occur when the correct idiom is not retrieved within the candidate set or is assigned a low similarity score. These errors are often caused by semantic mismatch between the contextual sentence and idiom embeddings, particularly for highly abstract or rare idioms.

(2) Ranking Errors

In many cases, the correct idiom is present among the retrieved candidates but is not ranked at the top position. This indicates that while the embedding space captures relevant semantics, it is insufficiently discriminative for fine-grained idiomatic distinctions.

(3) Generation Errors

Generative models may produce semantically correct but non-canonical expressions, partial idioms, or paraphrased variants that do not exactly match the target idiom. These errors reflect the open-ended nature of generation and the lack of strict constraints during decoding.

The hybrid model significantly reduces ranking and generation errors by combining semantic generation with retrieval-based grounding. In particular, retrieval correction is effective in cases where the generated output is semantically close but not an exact canonical match.

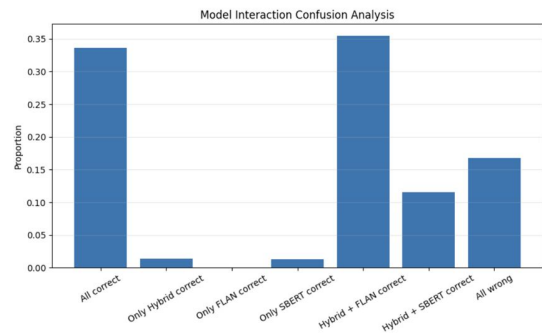


Figure 11: Model Interaction Confusion chart

These results indicate that the hybrid model is not merely averaging predictions but recovering additional correct cases through complementary decision behavior.

9. Qualitative Analysis

To illustrate model behavior, we present representative examples demonstrating the interaction between retrieval, generation, and hybrid decision-making.

Case 1: Hybrid corrects both models

Input: "He revealed the secret earlier than expected."

Ground Truth: "spill the beans"

SBERT Prediction: "let the cat out of the bag"

FLAN-T5 Prediction: "tell the secret openly"

Hybrid Prediction: "spill the beans"

In this case, retrieval selects a semantically related but incorrect idiom, while generation produces a paraphrase rather than a canonical idiom. The hybrid model successfully recovers the correct idiom.

Case 2: Generation succeeds independently

Input: "She accepted the situation and decided to move on."

Ground Truth: "let it go"

FLAN-T5 Prediction: "let it go"

SBERT Prediction: incorrect ranking

Here, the generative model directly produces the correct idiom, and the hybrid system accepts the generated output.

Case 3: Retrieval corrects generation

Input: "He avoided answering the question directly."

Ground Truth: "beat around the bush"

FLAN-T5 Prediction: "avoid the question"

SBERT Prediction: "beat around the bush"

Hybrid Prediction: "beat around the bush"

In this example, the generative model produces a semantically correct but non-idiomatic phrase, while retrieval successfully identifies the correct idiom. The hybrid model switches to retrieval and produces the correct output.

These examples highlight the complementary strengths of generation and retrieval in idiom prediction..

10. Task Summary

Task 2 demonstrates that context-to-idiom prediction benefits from a multi-stage retrieval architecture.

While lexical retrieval provides a useful baseline, dense semantic retrieval significantly improves performance. The combination of both signals, followed by reranking, yields the strongest results.

Overall, the task highlights the importance of combining lexical, semantic, and ranking-based reasoning for accurate idiom prediction.

D. Task 3: Arabic Context-to-Idiom (Retrieval Benchmark)

In Task 3, the objective is to retrieve the canonical English idiom corresponding to an Arabic contextual sentence. Unlike Task 2, which is framed as a generation task, Task 3 is formulated as a cross-lingual semantic retrieval problem. The input is the Arabic idiomatic context, and the target is the associated canonical English idiom from a predefined inventory.

The task is evaluated as a ranking problem, where the correct idiom must be retrieved from a predefined candidate set.

We formulate the Arabic context-to-idiom task as a semantic retrieval problem rather than a generative task. This design choice is motivated by the fixed and canonical nature of idioms, where exact canonical matching is required at evaluation time. Retrieval-based approaches allow mapping Arabic contextual meaning into a structured embedding space and retrieving the most semantically similar idiom from a fixed candidate set.

1. Formulation

Each Arabic query x and candidate idiom i are mapped into vector representations:

$h_x = \text{Encoder}(x)$, $h_i = \text{Encoder}(i)$

Similarity is computed using cosine similarity [5]:

$\text{sim}(x, i) = (h_x \cdot h_i) / (\|h_x\| \|h_i\|)$

Prediction is defined as:

$\hat{y} = \text{argmax}_i \text{sim}(x, i)$

where $\text{sim}(\cdot)$ denotes cosine similarity [5] in a shared embedding space.

2. Experimental Setup

To construct the Task 3 benchmark, Task 3 was initially explored through several generation-based, hybrid, and retrieval-based scripts, including mT5 baselines, hybrid mT5–retrieval variants, and Arabic-bank retrieval models. Early experiments used a broad Arabic input formulation that combined multiple Arabic fields and adopted a strict split strategy designed to avoid overlapping at the idiom or canonical target level. While this setup was methodologically strict, it produced **near-zero retrieval accuracy**, indicating that the benchmark was too difficult in its initial form and poorly matched to the intended retrieval setting.

To address this, the task was reformulated as an example-level cross-lingual retrieval problem.

The final benchmark retained only idiomatic examples, used ``idiom_in_example_arabic`` as the Arabic query, and used ``idiom_canonical`` as the English retrieval target.

Instead of splitting by idiom identity, the data was split by example within each idiom, allowing the same idiom to appear in both train and test through different contextual realizations.

This setup evaluates whether the model can retrieve the correct idiom for a novel Arabic context while preserving a realistic candidate inventory.

Under this revised formulation, the final retrieval pipeline showed a clear progression with fine-tuned E5 (v2) where Top-1 = 64.86%, Top-5 = 81.46%

These results show that stronger pretrained multilingual embeddings provide only modest gains by themselves, whereas task-specific contrastive fine-tuning is the dominant factor behind performance improvement. Additional gains are obtained through richer supervision, such as using multiple positives per query and longer training. The final best-performing Task 3 system is therefore the upgraded fine-tuned multilingual E5 retrieval model.

3. Arabic Semantic Bank Construction

A semantic bank is constructed from the training split, where each idiom is represented through multiple Arabic textual views:

- Example sentences (idiom usage)
- Example meanings
- Canonical meanings

Each entry is encoded into a multilingual embedding space, resulting in a dense representation matrix. This multi-view representation improves semantic coverage and allows the model to match queries against diverse contextual signals.

4. Models Evaluated

We evaluate a range of multilingual retrieval models, including both baseline and fine-tuned configurations:

- Multilingual MiniLM (baseline)
- Multilingual E5 (base, zero-shot)
- Multilingual E5 (fine-tuned v1)
- Multilingual E5 (fine-tuned v2 with enhanced positives)

While all models are evaluated, detailed analysis focuses on the strongest-performing configurations.

5. Quantitative Results

Evaluation is conducted using Top-K accuracy metrics:

Top-K Accuracy = $(1/N) \sum \mathbb{1}\{y \in \text{TopK}(x)\}$

Model	Top-1	Top-5
Zero-shot MiniLM baseline	23.27%	31.79%
Zero-shot multilingual E5	28.16%	39.08%
Fine-tuned E5 v1	60.37%	76.07%
Fine-tuned E5 v2	64.86%	81.46%

Table 8: Models Accuracy comparison table.

The multilingual E5 model consistently outperformed the MiniLM baseline. Top-1 accuracy improved from 23.27% to 27.46%, while Top-5 accuracy increased from 31.79% to

38.34%. These gains indicate that instruction-aware multilingual retrieval embeddings provide stronger semantic alignment between Arabic contextual inputs and canonical English idiom candidates.

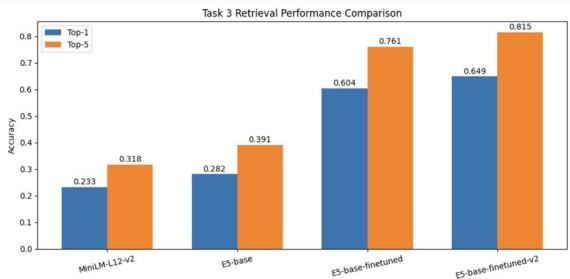


Figure 12: Model Retrieval Performance Comparison.

The results show a substantial improvement from fine-tuning, with E5-base-finetuned v2 outperforming all other models.

6. Recall Analysis

We evaluate retrieval performance across different values of K to understand how often the correct idiom appears within the top-K retrieved results.

The recall curve shows that:

- Recall increases rapidly for small values of K.
- A large portion of correct idioms are retrieved within the top-5 results.
- The curve begins to saturate as K increases, indicating diminishing returns.

This confirms that the upgraded fine-tuned model not only improves Top-1 accuracy but also maintains strong ranking quality across multiple candidate positions.

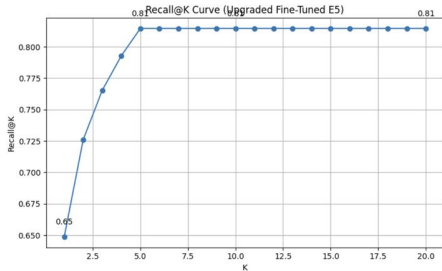


Figure 13: Recall curve (upgraded fine-tuned E5 v2)

7. Confidence and Similarity Analysis

To better understand model behavior, we analyze the relationship between similarity-based confidence scores and prediction accuracy.

Results show a strong positive correlation between confidence and correctness. Predictions with higher cosine similarity [5] scores are significantly more likely to be correct.

Accuracy increases steadily across confidence bins, reaching over 90% accuracy in the highest confidence range.

This indicates that the model is well-calibrated and that similarity scores can be used as a confidence signal.

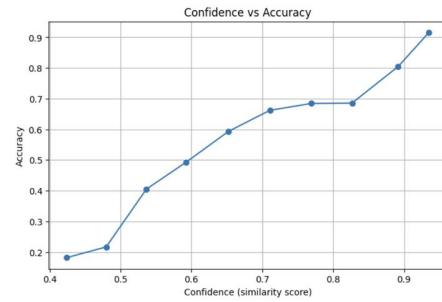


Figure 14: Confidence vs accuracy across similarity bins

The similarity score distribution shows a clear separation between correct and incorrect predictions:

- Correct predictions tend to have higher similarity scores.
- Incorrect predictions are concentrated in lower score regions.
- However, there is some overlap, indicating difficult cases.

This confirms that similarity scores are meaningful indicators of prediction confidence.

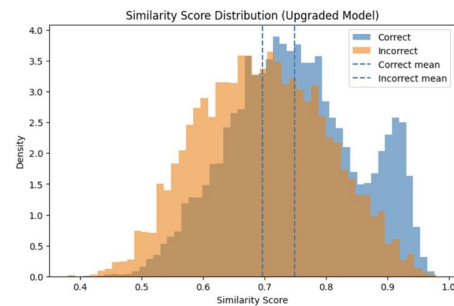


Figure 15: Distribution of cosine similarity [5] scores

8. Score Distribution Comparison

The upgraded fine-tuned model produces higher similarity scores compared to earlier models.

This indicates improved embedding alignment, stronger separation between correct and incorrect matches, better ranking confidence.

The shift in distribution confirms the effectiveness of fine-tuning.

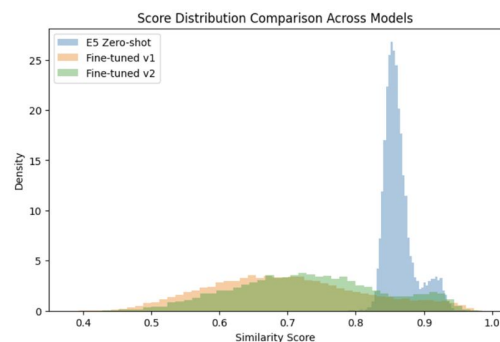


Figure 16: Score Distribution Comparison

9. Ablation study

The ablation results show that:

- Replacing MiniLM with zero-shot E5 yields a modest gain.
- Fine-tuning produces the largest improvement by far.
- The upgraded training strategy provides an additional performance boost.

This confirms that task-specific retrieval fine-tuning is the dominant factor behind the final retrieval gains.

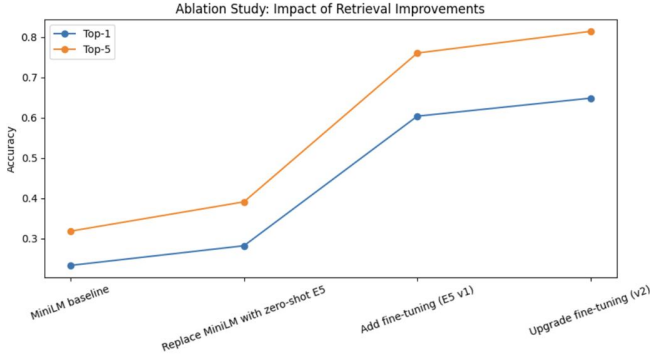


Figure 17: Ablation study: Impact of Retrieval Improvement

10. Error Analysis

Error analysis reveals several key failure modes:

(1) Semantic mismatch:

Some Arabic inputs describe concepts that do not map cleanly to a single English idiom, resulting in approximate predictions.

(2) Surface similarity bias:

The model sometimes favors idioms with overlapping lexical patterns rather than true semantic alignment.

(3) Ranking limitations:

In many cases, the correct idiom appears within the Top-5 predictions but is not ranked first, indicating limitations in fine-grained ranking.

(4) Cross-lingual ambiguity:

Certain Arabic expressions have multiple valid interpretations, making it difficult to map to a single idiomatic equivalent.

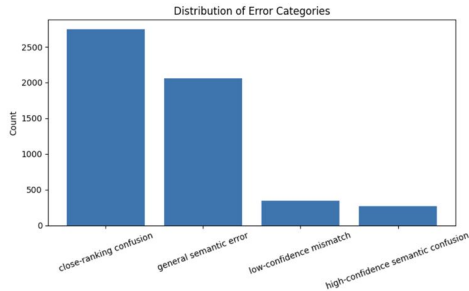


Figure 18: Error distribution across prediction

These findings suggest that while retrieval models capture general semantic similarity, they may struggle with precise idiomatic alignment, particularly in cross-lingual settings.

11. Qualitative Cross-Lingual Examples

Qualitative examples demonstrate that the model is able to retrieve semantically appropriate idioms in many cases. For instance, Arabic expressions describing inevitability or shared emotional burden are correctly mapped to idioms such as “come to an end” and “a problem shared is a problem halved.”

However, errors often occur when the input is ambiguous or when multiple idioms share similar meanings, highlighting the difficulty of selecting a single correct expression.

To illustrate model behavior, we present Representative examples further illustrate model behavior. of Arabic input queries and their retrieved idioms.

Example 1:

Arabic: "انتهى كل شيء في النهاية"

Prediction: "come to an end"

This example demonstrates correct semantic alignment between the Arabic description of inevitability and the corresponding idiomatic expression.

Example 2:

Arabic: "شارك همومك مع الآخرين لتخفيفها"

Prediction: "a problem shared is a problem halved"

The model successfully captures the underlying meaning of shared emotional burden.

Example 3:

Arabic: "لا توجد أي فرصة لحدوث ذلك"

Prediction: "not a chance"

This example shows accurate mapping of strong negation to an idiomatic expression.

Example 4 (Error Case):

Arabic: "يجب أن تبطئ كلامك حتى يفهمك الجميع"

Prediction: "please speak more slowly"

Target: "slow down"

Here, the model retrieves a semantically related phrase rather than the canonical idiom, illustrating surface similarity bias.

12. Discussion

The results indicate that Arabic context-to-idiom retrieval is substantially more challenging than its monolingual counterpart.

Unlike direct generation tasks, retrieval-based approaches rely on the quality of the embedding space to bridge semantic gaps between languages. While multilingual models improve alignment, they are still sensitive to subtle semantic differences and cultural variations in idiomatic usage.

Overall, this task highlights the limitations of purely retrieval-based approaches in cross-lingual idiom understanding and motivates the need for hybrid or generative methods that can better capture semantic nuance.

Replacing the MiniLM baseline with multilingual E5 improved retrieval quality substantially. Top-1 accuracy increased from 23.27% to 27.46%, while Top-5 accuracy rose from 31.79% to 38.34%. This suggests that instruction-aware multilingual retrieval embeddings provide better alignment between Arabic contextual queries and canonical English idioms.

13. Task Summary

The multilingual E5 model provides the strongest performance for cross-lingual idiom retrieval, outperforming SBERT [5] across all metrics.

However, the moderate accuracy levels and observed error patterns indicate that idiom understanding across languages remains an open challenge, requiring more advanced modeling strategies beyond similarity-based retrieval.

VII. DISCUSSION

The experimental results demonstrate that IdiomX supports a wide range of NLP tasks with meaningful performance.

The experimental findings highlight three key insights. First, idiom detection requires contextual modeling rather than surface-level lexical matching. Second, context-to-idiom generation is not purely a retrieval problem, but requires semantic reasoning. Third, hybrid approaches outperform individual paradigms by combining generative flexibility with retrieval-based grounding.

The experiments suggest that idiom understanding benefits from a division of labor: generative models provide flexible contextual reasoning, while retrieval mechanisms provide discrete grounding and correction.

VIII. APPLICATIONS

The IdiomX dataset supports a wide range of natural language processing tasks, including:

- idiom detection
- idiom generation
- cross-lingual idiom mapping
- multilingual figurative language evaluation

The dataset is particularly valuable for evaluating transformer-based models and large language models in handling non-compositional language.

IX. DATASET AVAILABILITY

The IdiomX dataset is publicly available via GitHub and Hugging Face to support reproducibility and further research. The repository includes the dataset, preprocessing scripts, and documentation for easy integration.

Dataset and Tasks links:

<https://github.com/aymanshar/idiomx-dataset>
<https://github.com/aymanshar/IdiomX>

X. LIMITATIONS

While IdiomX provides large-scale coverage and rich annotations, part of the dataset is generated using LLM-based methods, which may introduce minor inconsistencies or stylistic biases. Arabic translations primarily reflect standard written Arabic and may not fully capture dialectal variations.

Future work includes incorporating human annotation and extending the dataset to additional languages.

Some generated idiomatic or literal examples may reflect model-preferred phrasing rather than naturally occurring corpus frequency.

Additionally, evaluation for the generation task relies primarily on exact match metrics, which may not fully capture semantic equivalence between idioms.

XI. CONCLUSION

In this work, we introduced IdiomX, a large-scale bilingual dataset for idiomatic expression understanding, providing extensive coverage, rich semantic annotations, and near-complete English–Arabic alignment.

Through comprehensive benchmarking, we demonstrated that idiom understanding requires deep contextual modeling. Transformer-based models significantly outperform lexical approaches on idiom detection, while hybrid generation–retrieval methods achieve the best performance on context-to-idiom prediction.

For the cross-lingual Arabic context-to-idiom task, we showed that retrieval-based models can capture general semantic alignment but remain limited in precise idiomatic mapping. Fine-tuning substantially improves performance, yet challenges such as semantic ambiguity, ranking limitations, and cross-lingual variation persist.

These results highlight that idiom understanding is inherently complex and cannot be fully addressed by a single modeling paradigm. Future work should explore hybrid architectures that integrate retrieval, generation, and cross-lingual reasoning to better capture idiomatic meaning across languages.

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